# OPTICAL LOSS BUDGET STANDARD

Standard OSP-207 | Revision E | Effective 20 February 2026 | Owner: Outside Plant Engineering | Supersedes Revision D

## 1. Scope

This standard sets the component loss values, the received-power acceptance thresholds, and the remediation priority rules used to verify Gigabit Passive Optical Network (GPON) distribution plant. It applies at the 1490 nm downstream wavelength. All fibre in the access plant is ITU-T G.652.D single mode. Revision E withdraws the nominal 1:8 splitter allowance and introduces the SLA credit exposure rule in section 8.

## 2. Component loss values

These are design values. A component whose measured loss exceeds its design value is out of specification and shall be recorded as such.

| Component | Design loss | Unit |
|---|---|---|
| Single-mode fibre attenuation at 1490 nm | 0.25 | dB per km |
| Fusion splice | 0.10 | dB each |
| Mechanical splice | 0.30 | dB each |
| Mated connector pair | 0.50 | dB each |
| 1:4 optical splitter, insertion loss | 7.20 | dB |
| 1:8 optical splitter, insertion loss | certified per port | see section 3 |

## 3. Primary splitter insertion loss

**No nominal insertion loss shall be assumed for a 1:8 primary splitter.** Each 1:8 splitter is individually certified at manufacture and its ports differ. The insertion loss that applies to a distribution leg is the certified loss of the specific hub port that feeds that leg, as recorded in the plant record for the hub. Using an averaged or nominal figure in place of the certified port value is a reportable budgeting error.

## 4. Transmit reference

Optical line terminal (OLT) transmit power at the passive optical network (PON) port is commissioned and held at **+2.50 dBm**. End-to-end path loss is the difference between that transmit power and the power received at the optical network terminal (ONT).

## 5. Received-power acceptance thresholds

Every in-service ONT is classified from its measured received power as set out below. Boundaries are inclusive as written.

| Classification | Measured ONT received power |
|---|---|
| PASS | greater than or equal to -25.00 dBm |
| MARGINAL | below -25.00 dBm and greater than or equal to -27.00 dBm |

| | |
|---|---|
| FAIL | below -27.00 dBm |

## 6. Measurement tolerance

Power meters and optical time-domain reflectometers (OTDRs) in field service carry a combined uncertainty of plus or minus 1.00 dB. A difference between measured loss and design loss of 1.00 dB or less is within tolerance and shall not be raised as a fault. A difference greater than 1.00 dB indicates excess loss physically present in the plant and shall be investigated.

## 7. OTDR test method for distribution legs

A distribution leg is tested by shooting an OTDR from the fibre distribution hub (FDH) output port toward the multiport service terminal (MST). The trace datum, 0.00 km, is the FDH output port. Because the MST houses a splitter, the trace terminates at the MST input connector: **it cannot see any fibre, connector or splice downstream of the MST, and that includes all subscriber drop cable.** Excess loss downstream of an MST is therefore invisible to this test and must be isolated from received-power measurements taken at the subscriber terminal. An OTDR reports an event by its distance from the datum and does not report whether a splice is fusion or mechanical; the splice type is established from the plant record, and the design value for that type is the one that applies.

## 8. Remediation priority

Faults are ranked for remediation by severity first, placing FAIL ahead of MARGINAL. Faults of equal severity are then ranked by **SLA credit exposure**, in descending order. Any remaining tie is broken by the lowest measured received power. SLA credit exposure is the sum of the service-tier weights of every subscriber affected by the fault. Subscriber headcount alone is not a priority measure: a single high-tier subscriber can outrank an entire leg of residential service.

| Service tier | SLA credit weight |
|---|---|
| Residential 1G | 1 |
| Business 2G | 3 |
| Business 10G Priority | 6 |

## 9. Change control

A field change notice supersedes the as-built plant record for the segment it names, with effect from its issue date. Where a field change notice is in force, the geometry, splice schedule and component counts it specifies replace those in the as-built record for that segment. All other segments continue to be governed by the as-built record. Loss budgets computed against superseded geometry are invalid.